Toroidal Manifolds and Recursive Neural Dynamics: Topological Limits, Entropy Constraints, and Observer Interface Implications in Grid Cell Population Geometry

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Abstract

Toroidal population geometry motivates models of stable neural representation, but geometry alone does not establish dynamical attraction. We compare retained grid cell analysis outputs with an explicit model of two coupled oscillators. The recording audit supports spatial tuning descriptions but identifies pooling across separate recordings, preventing a valid simultaneous population interpretation. In the model, two circular phases define a toroidal coordinate space. Independent rotations can be periodic or quasi periodic, while attractive coupling produces a stable phase locked circle when detuning is smaller than twice the coupling strength. An amplitude extension yields an attracting invariant torus through explicit normal contraction. Exact solutions and finite simulations verify these results within stated numerical limits. Observation models further show that contraction can preserve an exactly encoded variable while finite resolution removes its distinguishability. Autonomous dynamics under projection require an additional consistency condition. Together, these findings specify the dynamical and observational assumptions needed to connect neural geometry with information persistence, entropy, and continuity across scales.

1 Introduction

Grid cells provide a tractable setting for studying how neural populations represent continuous variables. Individual cells fire at locations arranged approximately on a triangular lattice [1]. Across a population, relative firing patterns can encode position within that repeating structure. This periodic representation motivates toroidal coordinates and raises a dynamical question: what maintains the representation when neural activity is perturbed?

Published recordings support both geometric and dynamical descriptions of grid cell populations. Gardner et al. recovered toroidal population topology within simultaneously recorded modules, including across waking and sleep conditions [2]. Hermansen et al. found toroidal representations during one dimensional behavior [3]. Yoon et al. showed that relationships among simultaneous cells remained conserved despite changes in individual responses, supporting a low dimensional continuous attractor [4]. These findings motivate models in which nearby activity returns toward a continuous family of represented states [5].

The relation between spatial tuning and population dynamics nevertheless requires separate measurements. Di Sarra et al. found that timing perturbations could disrupt recovered topology while leaving spatial grid scores largely intact [6]. Their independent rate modulated Poisson model also produced toroidal signatures; a later correction replaces duplicated panels in the synthetic calibration figure [7]. A toroidal representation can thus be supported by several constructions, each with different implications for the underlying dynamics.

Coupled oscillators make these implications mathematically explicit. Neural rhythms and cross frequency interactions motivate phase descriptions [8–10]. Two independently specifiable circular phases have coordinate space T^2, while their equations determine which trajectories occur there. A dynamical attractor additionally requires recovery from nearby states in a specified ambient space. Information persistence adds another requirement: a variable must remain distinguishable under the available observation.

We examine these requirements through a recording audit, analytic oscillator models, and finite numerical checks. The central result is that toroidal coordinates, normal attraction, and information retention require distinct assumptions. The retained spatial maps do not establish a neural attractor. The oscillator model identifies a sufficient mechanism for attraction and shows how its interpretation depends on coupling and observation. These calculations provide testable conditions for a neural model rather than a fitted biological mechanism.

Section 2 introduces the data, equations, and verification methods. Section 3 presents the principal findings, followed by their interpretation in Section 4 and the conclusion in Section 5. Appendices A–H contain the derivations, processing details, numerical checks, and extended models.

2 Methods

2.1 Recording Audit and Observables

The empirical analysis examines five retained recording files from the Hafting data collection [1]. We inspect timestamps, cell labels, rate maps, persistence diagrams, and the transformations connecting them. Recording boundaries are preserved unless acquisition metadata establishes simultaneity. Reconstructing the earlier pooling rule allows its output to be checked against the stored population matrix. The audit also distinguishes retained numerical outputs from figures and calculations reported only in an annotated draft.

Spatial entropy is evaluated from the normalized rates across observed map bins. This defines a categorical statistic of spatial tuning. Topology is assessed by inspecting retained persistence outputs and their provenance. No calibrated neural H_2 reconstruction is performed in this study. Appendix C gives the recording inventory, processing parameters, entropy definition, and control checks needed to interpret these measurements.

2.2 Oscillator Model

The model represents each oscillator by a phase $\phi_i \in \mathbb{R}/2\pi\mathbb{Z}$ with constant natural frequency ω_i. Their joint coordinate space is $T^2 = S^1 \times S^1$. We use symmetric sinusoidal coupling [11, 12],

$$\begin{aligned} \dot{\phi}_1 &= \omega_1 + \epsilon \sin(\phi_2 - \phi_1), \\ \dot{\phi}_2 &= \omega_2 + \epsilon \sin(\phi_1 - \phi_2), \qquad \epsilon \geq 0. \end{aligned} \tag{1}$$

Here $\epsilon = 0$ gives independent rotations. With phase difference $\delta = \phi_2 - \phi_1$ and detuning $\Delta\omega = \omega_2 - \omega_1$, the relative dynamics reduce to

$$\dot{\delta} = \Delta\omega - 2\epsilon \sin \delta. \tag{2}$$

The lifted common phase advances at $(\omega_1 + \omega_2)/2$. This reduction identifies synchronization without assuming attraction transverse to the torus.

Normal attraction is tested by adding positive amplitude coordinates r_1, r_2 with

$$\dot{r}_i = -\lambda(r_i - 1), \qquad \lambda > 0. \tag{3}$$

The full state is embedded as $(r_1 \cos\phi_1, r_1 \sin\phi_1, r_2 \cos\phi_2, r_2 \sin\phi_2)$ in $\mathbb{R}^4$. This extension specifies the directions in which states can approach the unit product torus. Independent rotations, phase locking, radial perturbations, and a solvable forcing perturbation are compared under the same model framework. Appendices A and B provide the derivations and stability conditions.

2.3 Numerical and Observation Checks

Finite simulations test the analytic predictions in arbitrary model units. The principal frequencies are $(1, \sqrt{2})$, with coupling strengths 0, 0.1, and 0.6; a rational frequency control uses $(1, 3/2)$. Fourth order Runge–Kutta trajectories are compared with exact solutions and adaptive integration. Phase kicks and radial displacements test recovery

in different directions. Step refinement measures error over a fixed interval; Appendix D specifies all parameters, tolerances, and convergence limits.

Information is evaluated using specified observation maps. An exactly encoded binary variable and its finite resolution observation test whether contraction preserves distinguishability. Projections of the coupled phase model test whether an observed variable has autonomous dynamics. These examples use constructed states and observation rules; their definitions and calculations are in Appendix E.

3 Results

3.1 What the Retained Recordings Establish

The recording audit supports a description of spatial tuning but does not establish simultaneous toroidal population dynamics. The five files contain 23 cell recordings and 13 distinct labels. The stored $(59{,}992, 13)$ population matrix is reproduced exactly by merging repeated labels across separate recordings on one recording's time axis. Its embeddings consequently cannot establish the geometry or attraction of a simultaneous population. The 13 retained rate maps also do not reconstruct temporal population trajectories.

The retained topology outputs provide no calibrated evidence for a toroidal neural attractor. They contain H_1 intervals computed from the pooled matrix, while no matching H_2 output is retained. Further reconstructions appear in the annotated draft, but the supplied artifacts do not allow their independent reproduction. The spatial maps have mean entropy 10.860 bits under the stated rate normalization, compared with a 2,500-bin uniform maximum of 11.288 bits. This statistic describes spatial rate allocation. It does not measure memory duration or thermodynamic entropy production. Appendix C records the full numerical and provenance findings.

3.2 Synchronization and Normal Attraction

The same toroidal phase coordinates support different orbit structures. With independent nonzero frequencies, a rational frequency ratio gives periodic orbits and an irrational ratio gives dense quasi periodic orbits. Both phase directions remain neutral under perturbations. Under positive coupling, stable locking occurs when $|\Delta\omega| < 2\epsilon$, with

$$\delta_* = \arcsin\left(\frac{\Delta\omega}{2\epsilon}\right), \qquad \lambda_\delta = -\sqrt{4\epsilon^2 - (\Delta\omega)^2} < 0. \tag{4}$$

The stable locked set is a circle of fixed phase difference. It is a periodic orbit when the common frequency is nonzero and a circle of equilibria otherwise. The common phase remains neutral. At the equality threshold the equilibrium is nonhyperbolic; above the threshold the relative phase drifts. The complete branch and threshold analysis is given in Appendix A.

The amplitude extension establishes attraction to an invariant torus by an exact distance law. For $M = \{r_1 = r_2 = 1\}$,

$$d(z(t), M) = e^{-\lambda t} d(z(0), M). \tag{5}$$

Changing the sign of λ changes normal stability while preserving the flow on M. Under phase locking, M remains invariant and normally attracting even though typical limiting motion lies on a circle within it. A sufficiently strong normal contraction also gives persistence under small autonomous perturbations, subject to normal hyperbolicity [13]. This persistence concerns the manifold; preserving quasi periodic internal motion requires further hypotheses [14].

The finite calculations agree with the predicted locking, drift, and recovery. For $(\omega_1, \omega_2) = (1, \sqrt{2})$ and $\epsilon = 0.6$, the stable phase difference is approximately 0.3524 rad and its exponent is -1.1262 per model time unit. At $\epsilon = 0.1$, the mean slip rate is 0.3627 rad per model time unit. Phase kicks persist for independent rotations, whereas locking restores relative phase and retains a common offset. The radial examples recover

the predicted normal contraction. Fixed-interval refinement gives approximately fourth order convergence before the numerical floor. These results verify the finite model calculations; they supply no new biological measurements. Figure 2 and the error analysis are in Appendix D.

3.3 Information Persistence and Coarse Dynamics

Normal contraction does not by itself determine the recoverability of an encoded variable. In the binary example, two distinct initial displacements remain distinguishable at every finite time under exact observation. A fixed finite resolution eventually maps both to the same observed bin. Information loss thus depends on the encoding and measurement, even when the dynamical contraction is unchanged. Differential entropy in the normal coordinates decreases under contraction, but that change does not specify a thermodynamic entropy flux.

An observation supports autonomous coarse dynamics only if its evolution is consistent across states with the same observation. For fine and coarse flows F_t, G_t, a continuous surjection π, and a time conversion $c > 0$, the required relation is

$$\pi \circ F_t = G_{ct} \circ \pi. \tag{6}$$

The single phase ϕ_1 fails this test when coupling is nonzero: its velocity depends on the unobserved phase. The circular sum $(\phi_1 + \phi_2) \pmod{2\pi}$ instead follows an exact autonomous rotation. Appendix E derives the consistency condition and gives both examples. Continuity across descriptions therefore requires a compatible map and dynamics.

4 Discussion

4.1 From Population Geometry to Neural Dynamics

The results identify the extra evidence needed to connect a toroidal representation with a neural attractor. Published population studies support geometric organization and, in the

case of Yoon et al., a dynamical attractor interpretation for their recordings. The present audit limits what can be inferred from this repository's retained outputs. Its pooling error does not refute the published evidence. The explicit amplitude model shows why geometry and on-manifold trajectories alone leave normal stability undetermined.

A direct recovery experiment would connect these descriptions by measuring motion transverse to an independently estimated manifold. Simultaneous recordings should identify the population and module, control behavioral input, and resolve the relevant temporal scale. Displaced states could then be tested for normal contraction while allowing motion along the manifold. Return toward a manifold must be distinguished from return to the same timed trajectory. Topological controls would separately test the effects of noise, sampling, and preprocessing on the recovered shape. Appendix C states the associated sampling and calibration limits.

The oscillator results also distinguish synchronization from a two dimensional attracting representation. Coupling can reduce typical limiting motion to a circle while leaving the coordinate torus intact. An attracting torus in a larger space may coexist with this internal locking. A neural interpretation therefore needs a specified state space, a recovery measurement, and a description of the motion that remains on the recovered set.

4.2 Entropy, Observation, and Continuity

Information persistence requires both dynamics and an observation rule. Exact smooth contraction can preserve distinguishable states, whereas finite resolution can erase their observed difference. A report of decreasing entropy must accordingly identify its variable, normalization, and units. Spatial Shannon entropy, differential entropy, description length, and thermodynamic entropy describe different quantities. The thermodynamic balance in Appendix E requires a physical boundary and measured exchanges before it can constrain neural energetic cost.

Recursive processing inherits the distinctions preserved by its inputs. Repeating a deterministic operation on the same observation cannot recover distinctions already re-

moved by that observation. New inputs can supply additional information. Predictive coding and Minimum Description Length motivate conditional compression models [15–17], but do not establish monotonic compression through every biological layer. Likewise, Eq. (6) gives a condition to test when relating scales; oscillation or shared topology alone does not satisfy it.

Broader interpretations require additional models and measurements. Recurrence does not establish a neural self model, and toroidal spatial coding does not determine the topology of subjective experience. The present results concern state geometry, stability, and observation. Appendix F examines quaternion operators as a separate mathematical extension, and Appendix G states the physical assumptions needed for comparisons across frequency scales.

5 Conclusion

Toroidal phase coordinates specify a space of states, while the equations determine its orbit structure and stability. The coupled model produces independent rotations, phase locking onto a circle, and, with explicit normal contraction, an attracting invariant torus. Exact solutions and finite numerical checks support these model results. The retained neural artifacts establish spatial tuning descriptions but do not establish torus attraction.

The same distinction extends to information and scale. Contraction can preserve exact information while finite resolution removes it, and a coarse observation has autonomous dynamics only under a compatible projection. Connecting these mathematical results to neural persistence requires simultaneous recordings, recovery measurements, and explicit observation and entropy definitions.

A Geometry and Phase Dynamics

A.1 Periodic Spatial Coordinates

An idealized grid code has toroidal coordinates because positions repeat under lattice translations. Let $\Lambda = \mathbb{Z}\mathbf{e}_1 + \mathbb{Z}\mathbf{e}_2$ be a rank two lattice. Equivalent positions differ by an element of Λ, and the quotient $\mathbb{R}^2/\Lambda$ is a torus. Writing $\mathbf{x} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ gives circular coordinates $\phi_i = 2\pi u_i \pmod{2\pi}$. Figure 1 illustrates this construction. The lattice coefficients are wrapped; Cartesian coordinates are not generally independent periods of the same length.

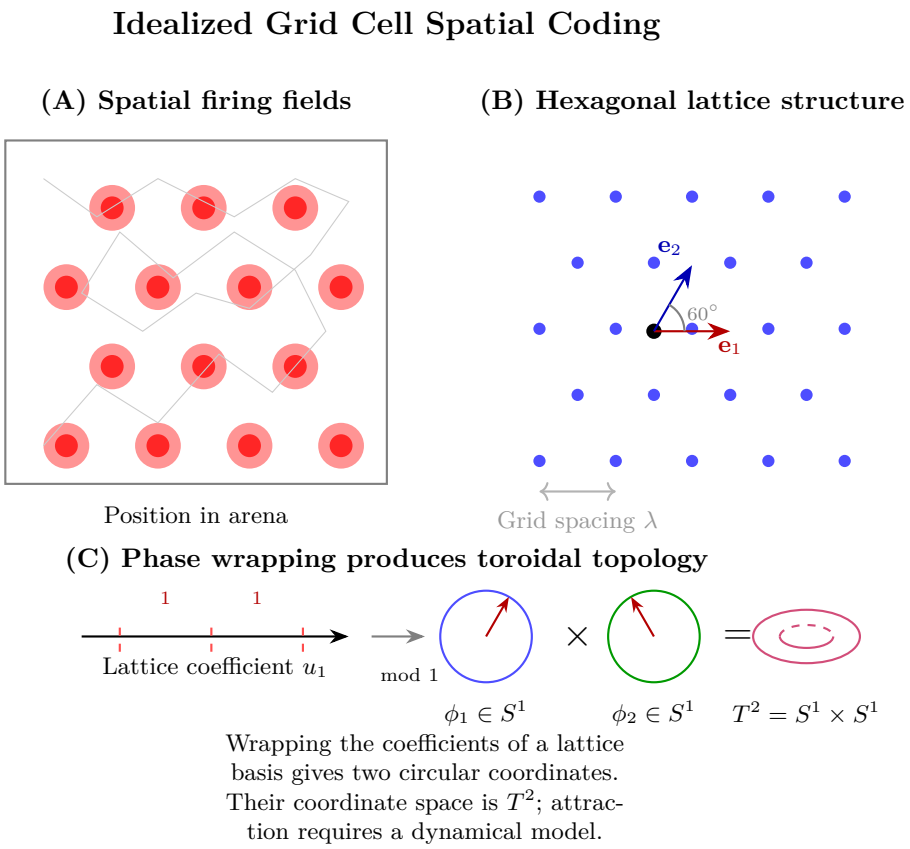


Figure 1: Idealized periodic spatial coding. (A) Schematic firing fields, not measured rate maps. (B) Two lattice basis vectors separated by 60°. (C) Wrapping their coefficients modulo one gives two circular coordinates and the quotient T^2.

A population firing map requires further assumptions to represent this quotient faithfully. It must separate distinct lattice phases and sample them adequately. A scalar firing rate from one neuron need not distinguish all phases. Spatial periodicity therefore

defines candidate coordinates without establishing an embedded population manifold or its dynamical stability.

Even a torus-compatible Betti signature does not uniquely determine a general space. Over a field, T^2 has $(\beta_0, \beta_1, \beta_2) = (1, 2, 1)$. The wedge $S^1 \vee S^1 \vee S^2$ has the same ranks. Additional assumptions or invariants are needed to identify the space. An unresolved H_2 feature is inconclusive about the underlying topology and says nothing by itself about attraction.

A.2 Circular Coordinates

A phase coordinate records position within an oscillation cycle [18, 19]. Consider a single idealized oscillator with phase variable $\theta_i(t)$ evolving as

$$\theta_i(t) = \omega_i t + \theta_{i,0}, \tag{A.1}$$

where $\theta_i(t)$ is the instantaneous phase of the i-th oscillator, ω_i is its angular frequency, and $\theta_{i,0}$ is the initial phase. The phase is identified modulo 2π, so $e^{i\theta_i(t)}$ lies on S^1. For $\omega_i \neq 0$ it traverses that circle periodically; $\omega_i = 0$ gives a fixed phase.

Two independently specifiable phases define a product space. When oscillators with phases $\theta_1(t)$ and $\theta_2(t)$ are considered jointly, each evolving on its own S^1, the state space of the combined system is the product manifold

$$T^2 = S^1 \times S^1, \tag{A.2}$$

which is the two-dimensional torus. This is the space of possible phase pairs; a particular trajectory need not visit all of it. More generally, n independently specifiable circular coordinates define $T^n = (S^1)^n$ [19]. Constraints or locking can restrict the accessible or limiting set.

The torus T^2 can be embedded in three-dimensional Euclidean space via the standard parametric representation:

$$x(\theta, \phi) = (R + r\cos\theta)\cos\phi, \tag{A.3}$$

$$y(\theta, \phi) = (R + r\cos\theta)\sin\phi, \tag{A.4}$$

$$z(\theta, \phi) = r\sin\theta, \tag{A.5}$$

For $R > r > 0$, this is an embedding of the abstract torus, not an inferred neural geometry. Here R is the major radius, defined as the distance from the center of the torus to the center of the tube. The minor radius r is the radius of the tube itself. The angles $\theta, \phi \in [0, 2\pi)$ correspond to the two independent phase variables.

A.3 Coupled Oscillator Dynamics on the Torus

Let $\phi_1, \phi_2 \in \mathbb{R}/2\pi\mathbb{Z}$. For C^1 coupling functions periodic in each phase, the equations

$$\dot{\phi}_1 = \omega_1 + \epsilon f_1(\phi_1, \phi_2), \tag{A.6}$$

$$\dot{\phi}_2 = \omega_2 + \epsilon f_2(\phi_1, \phi_2) \tag{A.7}$$

define a flow on T^2. They do not specify directions transverse to T^2 in a larger state space. Consequently, they cannot establish normal attraction of a neural manifold.

For $\epsilon = 0$ and nonzero frequencies, a rational ratio ω_1/ω_2 gives closed periodic trajectories; an irrational ratio gives trajectories dense in T^2. Both directions are neutral under phase perturbations. These rationality statements apply to constant independent rotations, not to arbitrary coupling functions or merely to the ratio of natural frequencies in a coupled system. For n independent rotations, density requires rational independence of the frequency vector: $k \cdot \omega \neq 0$ for every nonzero $k \in \mathbb{Z}^n$.

A.4 Locking Branches and the Critical Threshold

Subtracting the two equations in Eq. (1) gives Eq. (2). Adding them gives $\dot{\psi} = (\omega_1+\omega_2)/2$ for the locally lifted phase $\psi = (\phi_1 + \phi_2)/2$. The average is used on a lift because it is not a globally single-valued circular coordinate.

For $\epsilon > 0$ and $|\Delta\omega| < 2\epsilon$, solving $\dot{\delta} = 0$ gives the branch δ_* in Eq. (4) and the second branch $\pi - \delta_*$ modulo 2π. Linearizing gives $\dot{\eta} = -2\epsilon \cos\delta_* \, \eta$ to first order. Since $\cos\delta_* > 0$, its exponent is $-\sqrt{4\epsilon^2 - (\Delta\omega)^2}$. The other branch has the opposite sign and is unstable. The common phase remains neutral. A nonzero common frequency makes the stable circle a periodic orbit; zero common frequency gives a circle of equilibria. At $|\Delta\omega| = 2\epsilon$ the exponent vanishes, so exponential stability does not follow.

At the positive threshold $\Delta\omega = 2\epsilon > 0$, an exact solution with $\delta(0) = 0$ is

$$\delta(t) = 2\arctan\left(\frac{\epsilon t}{1+\epsilon t}\right), \qquad \frac{\pi}{2} - \delta(t) \sim \frac{1}{\epsilon t}. \tag{A.8}$$

It follows by setting $u = \tan(\delta/2)$, so that $\dot{u} = \epsilon(u-1)^2$. This orbit approaches the critical phase difference algebraically from below. Perturbations just above $\pi/2$ initially move away before winding around the circle; the critical equilibrium is not Lyapunov stable. Thus a vanishing linear exponent is not evidence for stable exponential locking.

If $|\Delta\omega| > 2\epsilon$, the phase difference drifts. The mean unwrapped slip rate is

$$\langle\dot{\delta}\rangle = \operatorname{sgn}(\Delta\omega)\sqrt{(\Delta\omega)^2 - 4\epsilon^2}. \tag{A.9}$$

For positive detuning this follows by integrating one cycle, $T_\delta = \int_0^{2\pi} (\Delta\omega - 2\epsilon\sin u)^{-1}\, du = 2\pi/\sqrt{(\Delta\omega)^2 - 4\epsilon^2}$. Drift alone is not a proof of dense quasi periodic motion: the common and slip rotation rates can be commensurate.

The phase model describes motion on a torus but has no ambient normal directions. Appendix B verifies the normal dynamics added in Eq. (3).

B Normal Attraction and Perturbation Calculations

In a specified ambient state space, an attracting invariant torus M is invariant under the flow and has a neighborhood whose trajectories approach it in distance. This local manifold attraction need not imply that every trajectory explores M, or that M is a minimal or transitive attractor. A sufficient mechanism is contraction in normal directions dominating tangent growth in forward and backward time, with appropriate smoothness and compactness assumptions. Normal hyperbolicity gives persistence of the invariant manifold under sufficiently small autonomous C^1 perturbations of the vector field [13]; no frequency nonresonance assumption is needed for that manifold persistence statement.

Equation (3) extends the phase pair with positive radial coordinates r_1, r_2. In the embedding $(r_1 \cos\phi_1, r_1 \sin\phi_1, r_2 \cos\phi_2, r_2 \sin\phi_2) \in \mathbb{R}^4$, the set $M = \{r_1 = r_2 = 1\}$ is an invariant T^2 and

$$r_i(t) - 1 = (r_i(0) - 1)e^{-\lambda t}. \tag{B.1}$$

For positive radii, the nearest point on each unit circle has the same phase. Hence $d(z, M)^2 = (r_1 - 1)^2 + (r_2 - 1)^2$, which gives Eq. (5). This proves normal attraction in the specified model. Replacing $\lambda > 0$ by $\lambda = 0$ or $\lambda < 0$ leaves the flow on M unchanged but makes the normal directions neutral or locally repelling. For the repelling case, the polar description is used while the radii remain positive. Topology and on manifold trajectories alone therefore cannot determine normal stability.

B.1 Normal Hyperbolicity and Internal Motion

The normal attraction can coexist with phase locking. For independent rotations the tangent exponents are zero. For the sinusoidally coupled pair a sufficient domination bound is $\lambda > 2|\epsilon|$, because the tangent Jacobian has norm at most $2|\epsilon|$. Under locking the same torus remains invariant and normally attracting, but its typical limiting motion is the stable circle within it.

Explicitly, the phase Jacobian is

$$J(\phi) = \epsilon \cos\delta \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \|J(\phi)\|_2 \le 2|\epsilon|. \tag{B.2}$$

The forward and inverse tangent propagators are consequently bounded by $e^{2|\epsilon|t}$ for $t \ge 0$. Normal displacements decay as $e^{-\lambda t}$. The stated strict inequality makes normal contraction dominate both tangent bounds. Attraction in the explicit model holds for every $\lambda > 0$; this stronger sufficient bound supplies normal hyperbolicity.

B.2 A Solvable Perturbation

A controlled perturbation gives a second exactly checkable example. With independent phases, replace Eq. (3) by

$$\dot{r}_i = -\lambda(r_i - 1) + a \sin\phi_i. \tag{B.3}$$

The attracting invariant torus is now the graph

$$r_i^*(\phi_i) = 1 + \frac{a(\lambda \sin\phi_i - \omega_i \cos\phi_i)}{\lambda^2 + \omega_i^2}. \tag{B.4}$$

For $|a| < \min_i \sqrt{\lambda^2 + \omega_i^2}$ the graph has positive radii. Substitution gives $\frac{d}{dt}(r_i - r_i^*(\phi_i)) = -\lambda(r_i - r_i^*(\phi_i))$. The graph displacement therefore contracts exactly. Distance to the graph is bounded above by this displacement at the same phases, proving attraction, although distance to the original unit torus need not vanish. This is a special solvable perturbation, not a numerical proof for arbitrary perturbations.

B.3 Scope of Persistence Results

Persistence of a torus as a normally attracting manifold does not ensure that its internal flow remains quasi periodic. Generic perturbations can change rotation rates and introduce locking. Results preserving quasi periodic motion require the additional regularity, arithmetic, nondegeneracy, and parameter assumptions of the particular theorem [14].

Classical Hamiltonian KAM results [20–22] cannot simply be applied to a dissipative neural model. Physical energy dissipation alone establishes neither normal hyperbolicity nor those quasi periodic persistence hypotheses.

C Data Processing and Provenance

C.1 Recording Inventory and Processing

The repository retains five MAT files from the Hafting data collection [1]. We inspect position timestamps, coordinates, and cell timestamp arrays in each file. The audit records input hashes, cell labels, and matrix dimensions in `outputs/claim_audit/empirical_provenance.json`. Separate files are treated as separate recordings unless acquisition metadata establishes simultaneity.

The corrected activity builder bins one selected recording at a time. It uses complete 20 ms bins and divides spike counts by the bin width to obtain rates. Each empirical command requires a recording name and writes to a recording-specific output directory. We separately reconstruct the earlier pooling rule and compare its matrix entry by entry with the retained legacy matrix. This reconstruction identifies provenance; it is not used as a new biological population analysis.

Spatial rate maps use a 50×50 position grid. Spike counts and estimated occupancy time are smoothed separately with a Gaussian kernel of width two spatial bins. Their ratio is evaluated where smoothed occupancy exceeds 0.01 s. Occupancy is approximated by sample count times the median tracking interval. New map generation assigns spikes to the nearest retained position timestamp. Irregular tracking and long gaps require further treatment before quantitative biological interpretation.

C.2 Topology and Entropy Checks

The topology audit reads retained persistence diagrams and inspects the code that generated them. It counts finite H_1 intervals and records their lifetimes. It also checks whether

matching H_2 outputs and reported control implementations are present. We do not rerun or calibrate the full neural PCA, UMAP, and persistent homology pipeline in this audit.

Spatial entropy provides a separate description of the retained rate maps. For finite nonnegative rates r_b with positive total, we use

$$p_b = \frac{r_b}{\sum_j r_j}, \qquad H = -\sum_b p_b \log_2 p_b, \tag{C.1}$$

with $0 \log_2 0 = 0$. This weights spatial bins by rate, rather than by occupancy-weighted spike probability. The normalization must therefore accompany any interpretation of the statistic.

The other entropy routines use different categorical variables. The population routine normalizes activity across neuron labels within each time bin. It measures activity-share entropy conditional on positive total activity. The temporal routine partitions samples into contiguous windows and measures the allocation of activity among those windows. Neither routine estimates a process entropy rate or a memory lifetime. Silent or missing distributions are undefined; summaries report how many bins have defined values.

The annotated draft supplied for revision contains additional raster figures and numerical reports. We inspect those reports as part of the document record. Their source recordings, executable notebooks, and numerical arrays are not included in the materials available for this audit. Printed plots alone do not supply a reproducible reconstruction.

C.3 Detailed Audit Results

C.3.1 Recording Provenance

The five retained files contain 23 cell recordings and 13 distinct labels (Table 1). Some labels repeat between trials. These counts do not establish the number of distinct biological cells or a simultaneous pooled population.

Table 1: Retained recording files. Position counts are before removal of nonfinite coordinates. Distinct cell labels across files need not identify distinct biological cells.

File	**Position samples**	**Cell recordings**
`Hafting_Fig2c_Trial1.mat`	59,596	7
`Hafting_Fig2c_Trial2.mat`	59,723	7
`Hafting_Fig2d_Trial1.mat`	59,596	3
`Hafting_Fig2d_Trial2.mat`	59,723	3
`rat_10925.mat`	89,652	3

The legacy population matrix does not preserve recording boundaries. Its shape is $(59{,}992, 13)$, and the audit reproduces it exactly by merging matching cell labels across files on one selected recording's time axis. Consequently, its embeddings and population statistics cannot establish simultaneous neural dynamics. The corrected selection rule prevents this pooling.

The retained rate maps describe spatial tuning within the limits of their provenance. Earlier exports reused filenames for repeated cell labels, leaving 13 maps without a one-to-one record of all 23 cell recordings. Those maps do not reconstruct temporal population trajectories. Their geometry cannot establish normal attraction.

C.3.2 Persistent Homology and Control Reports

The retained legacy diagrams provide no calibrated toroidal population result. They contain 32 finite H_1 intervals for PCA and 84 for UMAP. The maximum lifetimes are 0.668947 and 4.001927 in the respective embedding metrics. These values describe stored calculations on the invalid pooled matrix. UMAP changes distances, so the two maxima are not directly comparable. No matching H_2 output is retained, and the topology entry point computes only H_0 and H_1.

The annotated draft documents further analyses but does not resolve their reproducibility. It displays rate maps, persistence barcodes, and a UMAP embedding attributed to Gardner modules. It also reports an H_2 calculation and a separate Colab parameter sweep. Some numerical values differ from the repository manuscript. We retain these as unverified draft reports, rather than selecting one version as a measured result. In

particular, neither a plotted persistence gap nor a stated threshold establishes a noise floor without calibration.

The retained ablation code does not implement the draft's reported timestamp-jitter experiments. It compares spatial smoothing, neuron subsets, and shuffling of binned activity columns. Its legacy outputs inherit the pooling error. Reordering a list of spike timestamps before histogramming leaves the event train unchanged; a temporal surrogate must alter event times or their relationships. Spatial smoothing, temporal smoothing, and PCA whitening also modify different quantities. Their effects cannot be interchanged in an interpretation.

C.3.3 Spatial Entropy

The retained spatial maps have mean entropy 10.860 bits under Eq. (C.1). A uniform distribution over all 2,500 bins has entropy $\log_2(2500) \approx 11.288$ bits. The earlier reported value of approximately 1.49 bits is not reproduced from these maps. Binning, smoothing, missing positions, and normalization affect the comparison.

The stored mean population entropy of 0.260240 bits belongs to the pooled matrix and has a different definition. It is not a valid simultaneous population statistic. Neither entropy value measures temporal information persistence or thermodynamic production. Appendix E examines those separate questions using explicit dynamical and observation models.

C.4 Sampling and Topological Calibration

Temporal sampling must match the frequency being tested. A 100 ms bin gives a sampling rate of 10 Hz and a Nyquist frequency of 5 Hz. It cannot resolve the full 4–8 Hz theta band. Gardner et al. used 10 ms spike count bins with separate smoothing and subsampling choices [2]. Sampling rate, filtering, and the target observable must be specified together.

Topological calibration is a separate requirement. The same neural preprocessing and filtration pipeline should be tested against known topology with matched noise and sam-

pling. Coverage, dependence, metric changes, and preprocessing can alter persistence. Dense sampling alone does not universally inflate spurious H_1 features. The finite oscillator calculations test dynamical equations and do not provide this topology calibration.

D Numerical Methods and Verification

D.1 Simulation Design

All simulations use arbitrary model units. No parameters are fitted to neural recordings. The script `scripts/oscillator_validation.py` uses $(\omega_1, \omega_2) = (1, \sqrt{2})$ and compares $\epsilon = 0$, 0.1 (drift), and 0.6 (locking), with a rational control $(1, 3/2)$ at $\epsilon = 0$. Initial phases are $(0.2, 0.9)$ and the fixed step fourth order Runge–Kutta (RK4) runs use $h = 0.02$ over 200 arbitrary time units. Radial runs use $\lambda = 2$, initial radii $(0.7, 1.3)$, and forcing $a = 0$ or 0.1 over 20 time units. The unforced radial case is also run under phase locking. A separate short control reverses the normal rate without changing the flow on the torus.

D.2 Analytic Benchmarks

The independent runs agree with the exact rotation solution. The rational control returns after 4π. Density for the irrational case follows analytically from the assumed exact frequency ratio; finite samples and floating point arithmetic do not prove it. A 0.2 radian phase kick persists under independent rotation. Locking restores the relative phase but retains a common phase offset.

The coupled runs agree with the locking and slip predictions. At $\epsilon = 0.6$, Eq. (4) gives $\delta_* \approx 0.3524$ rad and $\lambda_\delta \approx -1.1262$ per model time unit. At $\epsilon = 0.1$, Eq. (A.9) gives a mean slip rate of 0.3627. Full cycle crossing times and independent quadrature check the slip rate without endpoint averaging bias. The radial runs agree with the exact contraction and forced-graph solutions in Eqs. (3) and (B.4). Figure 2 compares these behaviors.

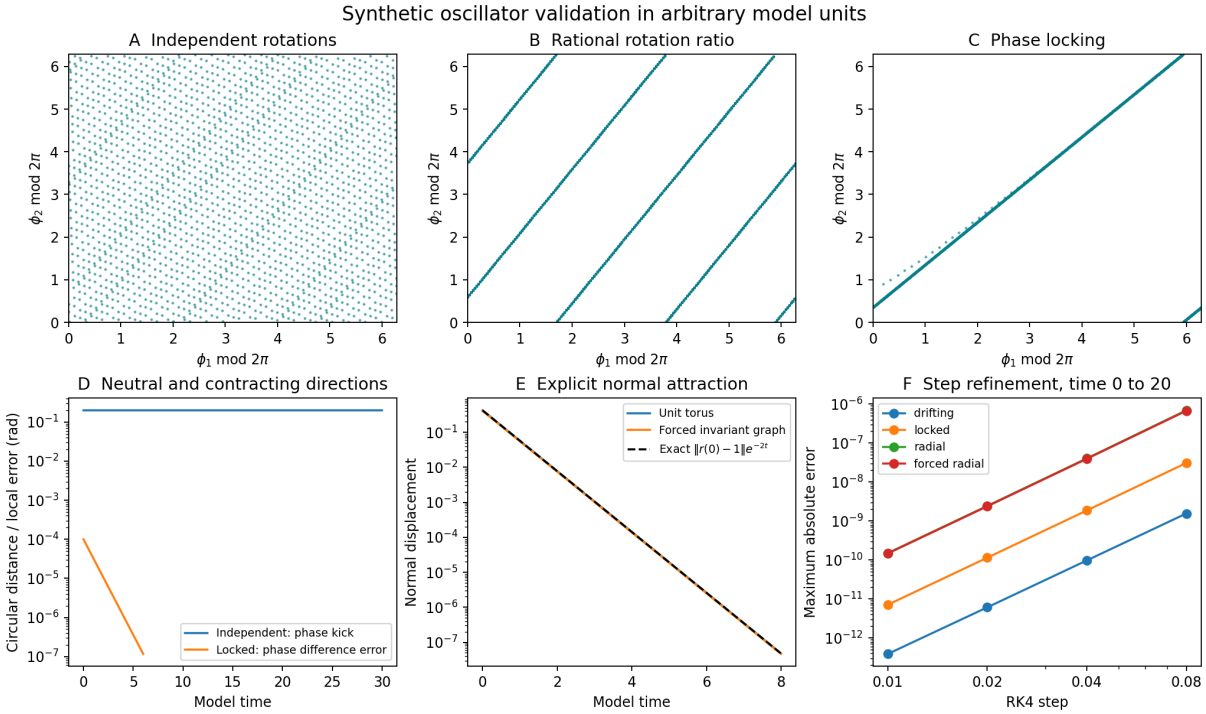


Figure 2: Finite synthetic oscillator calculations in arbitrary units. Phase plots use wrapped coordinates for display; integration uses unwrapped angles. Independent irrational rotations, rational rotations, and phase locking have the same phase coordinate space but different orbit structure. Perturbation checks distinguish neutral phase directions from contraction of phase difference and normal radial displacement. Step refinement checks numerical error against an adaptive reference. None of these panels is a neural recording or empirical attractor reconstruction.

D.3 Numerical Error and Limits

Step refinement measures numerical error over a fixed 20 unit interval. RK4 uses $h = 0.08, 0.04, 0.02, 0.01$. The adaptive DOP853 reference uses relative tolerance 10^{-12}, absolute tolerance 10^{-13}, and maximum step 0.1. A tighter DOP853 run and an independent RK45 integration check the locked case. The errors decrease at approximately fourth order before the numerical floor. At $h = 0.01$, maximum absolute errors against the reference are approximately 3.9×10^{-13} for drift, 7.1×10^{-12} for locking, and 1.5×10^{-10} for both radial cases. Tests also check exact solutions, stable and unstable branches, and neutral common phase. The CSV trajectories, convergence table, near threshold comparisons, environment versions, and JSON summary are retained with the script.

The supplementary script `scripts/information_validation.py` checks the critical solution in Eq. (A.8) against adaptive integration and the observation examples in Appendix E.2. These calculations also use synthetic parameters and do not estimate biological memory times.

The critical solution is evaluated through $u/(1+u)$, with $u = \epsilon t$, to avoid cancellation near $t = 0$. Quantized bin labels must remain finite and below 2^{53}, where consecutive integers are representable in double precision. Requests beyond that range are rejected rather than interpreted as loss of information. Floating point rounding near bin boundaries and underflow at very long times remain numerical limits of the observation calculation.

Finite agreement checks the implementation for the stated models and intervals. Near the locking threshold, slow drift or recovery can be misclassified over short windows.

Fixed RK4 is conditionally stable: for the scalar normal mode its stability polynomial must satisfy $|1 - z + z^2/2 - z^3/6 + z^4/24| < 1$ with $z = \lambda h$; the tested maximum $z = 0.16$ lies within this interval. These checks do not establish accuracy for arbitrary steps or long time orbit coverage. Stochastic robustness and a neural noise floor are not tested.

E Observation Maps and Entropy Calculations

E.1 Observation Classes and Recursive Processing

An observation map specifies which distinctions a representation retains. Let $\pi : X \to O$ map system states to observations. Define $x \sim_\pi y$ when $\pi(x) = \pi(y)$. Equality makes this relation reflexive, symmetric, and transitive. Its class at x is $[x]_\pi = \{y : \pi(y) = \pi(x)\}$. A noninjective map identifies distinct states; finite dimension alone does not imply noninjectivity.

A deterministic downstream map cannot distinguish states already identified by π. If $R : O \to O$ and $\pi(x) = \pi(y)$, then $R^n(\pi(x)) = R^n(\pi(y))$ for every integer $n \geq 0$. This is a statement about processing the same observation. It does not prohibit new observations or external inputs from providing additional information.

Toroidal coordinates are one possible observation map. Given $\varphi_1, \varphi_2 : X \to S^1$, define $\Phi(x) = (\varphi_1(x), \varphi_2(x)) \in T^2$. Equality of both phase coordinates is equivalent to equality of the ordered pair. This product property alone gives no surjectivity, embedding, or attractor theorem.

E.2 Contraction and Information Persistence

Normal attraction controls distance to a manifold, not the recoverability of an encoded variable. Phase locking suppresses relative phase differences asymptotically, but uniqueness of a smooth ordinary differential equation prevents distinct exact states from merging at finite time on their common interval of existence. This mathematical injectivity need not survive measurement noise, finite resolution, an incomplete observation, or numerical rounding.

For example, encode an equiprobable binary label B in a radial displacement $x_B(0) \in \{0.25, 0.75\}$ and let $x_B(t) = e^{-2t}x_B(0)$. Exact observation distinguishes the labels at every finite time, so $I(B; x_B(t)) = 1$ bit. With the explicitly chosen observation $Q_\Delta(x) = \lfloor x/\Delta \rfloor$, both labels occupy bin zero once $t > \frac{1}{2}\log(0.75/\Delta)$. Their observed mutual information is then zero. For $\Delta = 0.1$ and 0.05, these times are approximately 1.0075 and 1.3540 model units. These are consequences of the chosen code and quantizer, not neural memory lifetimes or universal information decay rates.

Differential entropy describes yet another quantity. For a two dimensional normal displacement $X(t) = e^{-\lambda t}X(0)$ with a density and finite differential entropy, the change-of-variables formula gives

$$h(X(t)) = h(X(0)) + \log|\det(e^{-\lambda t}I_2)| = h(X(0)) - 2\lambda t, \tag{E.1}$$

in nats relative to fixed normal coordinates. This decreasing differential entropy coexists with finite-time invertibility. It is not thermodynamic entropy production; no heat flux follows without a physical model.

E.3 Consistency Across Scales

For a relation between fine and coarse descriptions, one could seek a continuous surjection π and flows F_t, G_t satisfying Eq. (6) for a specified time conversion $c > 0$. Such an autonomous coarse flow requires preservation of observation classes: $\pi(x) = \pi(y)$ must imply $\pi(F_t(x)) = \pi(F_t(y))$. For differentiable maps and vector fields f, g, differentiating at $t = 0$ gives the necessary condition $D\pi_x f(x) = c\, g(\pi(x))$. Establishing a continuous or smooth quotient flow also requires the corresponding topological or regularity assumptions.

The sine pair supplies a direct counterexample to automatic closure. With $\pi(\phi_1, \phi_2) = \phi_1 \pmod{2\pi}$ and $\epsilon \neq 0$, the states $(0, 0)$ and $(0, \pi/2)$ have identical observations but projected velocities ω_1 and $\omega_1 + \epsilon$. Thus this observation does not define an autonomous one phase flow. In contrast, $\sigma = (\phi_1 + \phi_2) \pmod{2\pi}$ satisfies $\dot{\sigma} = \omega_1 + \omega_2$ exactly. The supplementary script checks both examples. Neither these mathematical factors nor the grid cell maps establish continuity or information conservation across molecular, neural, and behavioral scales.

E.4 Entropy and Physical Cost

Thermodynamic entropy requires a physical system boundary and specified exchanges. In units of k_B, its balance is [23]

$$\dot{S} = \dot{S}_i + \dot{S}_e, \qquad \dot{S}_i \geq 0. \tag{E.2}$$

At steady state, $\dot{S}_e = -\dot{S}_i \leq 0$. Export is strictly positive in magnitude only when internal production is positive. This balance does not determine the dimension or topology of a neural manifold.

A proposed entropy funnel must specify the quantity it changes. A restriction of represented states can lower a chosen description length without determining heat export, Shannon entropy, or an entropy rate. Minimum Description Length and predictive cod-

ing models motivate conditional compression hypotheses [15–17]. They do not establish decreasing description length at every biological layer. Comparing the metabolic costs of different manifolds remains an open modeling and measurement problem.

F Quaternion Operators and Phase Coordinates

Three independently specifiable circular phases have coordinate space T^3. Unit quaternions instead form S^3, the double cover of $SO(3)$. No continuous embedding of T^3 into S^3 exists. Invariance of domain would make its image open, and compactness would make the image closed. Since S^3 is connected, such an embedding would be a homeomorphism. This contradicts $\pi_1(T^3) = \mathbb{Z}^3$ and $\pi_1(S^3) = 0$ [24].

A separately specified quaternion curve can describe rotation operators. For a pure quaternion angular velocity $\Omega(t)$,

$$\dot{q}(t) = \tfrac{1}{2}\Omega(t)q(t), \qquad |q(0)| = 1 \tag{F.1}$$

preserves $|q(t)| = 1$. This equation does not encode all phase triples or derive their dynamics from three natural frequencies. An action on phase states must be defined separately.

Ordinary phase translations commute. With $T_a(\phi) = \phi + a \pmod{2\pi}$, one has $T_a T_b = T_b T_a$. A noncommutative neural model therefore requires additional operators and a specified composition rule. Comparing $A(B(x))$ and $B(A(x))$ from matched states could test such a rule. Merely changing which frequency band is sampled first also changes its timing and can produce ordinary order dependence.

Pairwise coupling statistics do not specify this operator law. Phase amplitude coupling assigns different roles to a selected phase and amplitude [25]. The weighted phase lag index measures pairwise synchronization through the imaginary cross-spectrum [26]. Neither alone establishes quaternion composition. An empirical test would need controls for delays, common input, amplitudes, and omitted state variables.

G Physical Assumptions Across Frequency Scales

Electromagnetic descriptions require assumptions about sources and material response. In a vacuum region without charges or currents, Maxwell's equations imply [27]

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \partial_t^2 \mathbf{E} = 0. \tag{G.1}$$

For a homogeneous, isotropic, nondispersive Ohmic medium with constant μ, ϵ, σ, no free charge density, and no impressed current, the equation becomes

$$\nabla^2 \mathbf{E} - \mu\epsilon \partial_t^2 \mathbf{E} - \mu\sigma \partial_t \mathbf{E} = 0. \tag{G.2}$$

Neural tissue additionally requires source currents, boundary conditions, and appropriate dispersive properties. These equations do not supply a measured mechanism for toroidal neural dynamics.

Frequency comparisons do not establish coupling between scales. Relative to 10^3 Hz, the range 10^9–10^{13} Hz spans six to ten orders of magnitude. Channel gating times and inverse relaxation times are not automatically coherent oscillation frequencies. A model linking these scales would need measured variables and specified interactions.

The band 0.1–10 THz corresponds to photon energies 0.414–41.36 meV through $E = h\nu$. At 310 K, $k_B T \approx 26.7$ meV [28]. Part of that band is near a thermal energy scale, but the comparison does not establish coherence or equate all photon energies with hydrogen bond dissociation. Molecular spectroscopy and Fröhlich's coherence proposal motivate further questions [29, 30]. They do not demonstrate coupling to the grid cell model in this paper.

H Formalization and Reproducibility

The mathematical checks are analytic calculations and numerical tests, not completed machine proofs. The retained Lean sources contain proof placeholders, undefined identi-

fiers, and no pinned build. Declaring an abstract type named S^1 does not construct the geometric circle. The original prototypes remain in the repository for future development; the paper does not treat them as verification of topology, stability, or empirical evidence.

The repository contains the data audit, simulation scripts, tests, and dependency versions. Run `scripts/oscillator_validation.py` and `scripts/information_validation.py` for the synthetic calculations. Run `scripts/audit_empirical_provenance.py` for the retained-artifact inventory. The README gives the test command, manuscript build instructions, and recording-selection requirements. Repository address: https://github.com/ezernackchristopher97-cloud/ENTPTCCONPAPER26.

References

[1] Torkel Hafting, Marianne Fyhn, Sturla Molden, May-Britt Moser, and Edvard I. Moser. Microstructure of a spatial map in the entorhinal cortex. *Nature*, 436(7052): 801–806, 2005. doi: 10.1038/nature03721.

[2] Richard J. Gardner, Erik Hermansen, Marius Pachitariu, Yoram Burak, Nils A. Baas, Benjamin A. Dunn, May-Britt Moser, and Edvard I. Moser. Toroidal topology of population activity in grid cells. *Nature*, 602(7895):123–128, 2022. doi: 10.1038/s41586-021-04268-7.

[3] Erik Hermansen, David A. Klindt, and Benjamin A. Dunn. Uncovering 2-D toroidal representations in grid cell ensemble activity during 1-D behavior. *Nature Communications*, 15:5429, 2024. doi: 10.1038/s41467-024-49703-1.

[4] KiJung Yoon, Michael A. Buice, Caswell Barry, Robin Hayman, Neil Burgess, and Ila R. Fiete. Specific evidence of low-dimensional continuous attractor dynamics in grid cells. *Nature Neuroscience*, 16(8):1077–1084, 2013. doi: 10.1038/nn.3450.

[5] Si Wu, K. Y. Michael Wong, C. C. Alan Fung, Yuanyuan Mi, and Wenhao Zhang. Continuous attractor neural networks: candidate of a canonical model for neural information representation. *F1000Research*, 5:156, 2016. doi: 10.12688/f1000research.7387.1.

[6] Giovanni di Sarra, Siddharth Jha, and Yasser Roudi. The role of oscillations in grid cells' toroidal topology. *PLOS Computational Biology*, 21(1):e1012776, 2025. doi: 10.1371/journal.pcbi.1012776.

[7] Giovanni di Sarra, Siddharth Jha, and Yasser Roudi. Correction: The role of oscillations in grid cells' toroidal topology. *PLOS Computational Biology*, 21(10):e1013623, 2025. doi: 10.1371/journal.pcbi.1013623.

[8] György Buzsáki. *Rhythms of the Brain*. Oxford University Press, New York, 2006.

[9] Ryan T. Canolty, Erik Edwards, Sarang S. Dalal, Maryam Soltani, Srikantan S. Nagarajan, Heidi E. Kirsch, Mitchel S. Berger, Nicholas M. Barbaro, and Robert T. Knight. High gamma power is phase-locked to theta oscillations in human neocortex.

Science, 313(5793):1626–1628, 2006. doi: 10.1126/science.1128115.

[10] Adriano B. L. Tort, Robert W. Komorowski, Joseph R. Manns, Nancy J. Kopell, and Howard Eichenbaum. Theta-gamma coupling increases during the learning of item-context associations. *Proceedings of the National Academy of Sciences*, 106(49): 20942–20947, 2009. doi: 10.1073/pnas.0911331106.

[11] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, Berlin, 1984.

[12] Steven H. Strogatz. From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D*, 143(1–4):1–20, 2000. doi: 10.1016/S0167-2789(00)00094-4.

[13] Pierre Berger and Abed Bounemoura. A geometrical proof of the persistence of normally hyperbolic submanifolds. *Dynamical Systems*, 28(4):567–581, 2013. doi: 10.1080/14689367.2013.835386.

[14] Hendrik W. Broer, George B. Huitema, and Mikhail B. Sevryuk. *Quasi-Periodic Motions in Families of Dynamical Systems*, volume 1645 of *Lecture Notes in Mathematics*. Springer, 1996. doi: 10.1007/978-3-540-49613-7.

[15] Rajesh P. N. Rao and Dana H. Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1):79–87, 1999. doi: 10.1038/4580.

[16] Ming Li and Paul Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications*. Springer, 3rd edition, 2008.

[17] Benjamin Prada, Shion Matsumoto, Abdul Malik Zekri, and Ankur Mali. Bridging predictive coding and MDL: A two-part code framework for deep learning, 2025. URL https://arxiv.org/abs/2505.14635v2. Preprint, version 2.

[18] Arthur T. Winfree. *The Geometry of Biological Time*, volume 12 of *Interdisciplinary Applied Mathematics*. Springer, 2nd edition, 2001. doi: 10.1007/978-1-4757-3484-3.

[19] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press, 2nd edition, 2015.

[20] Andrey N. Kolmogorov. On conservation of conditionally periodic motions under small perturbations of the Hamiltonian. *Doklady Akademii Nauk SSSR*, 98:527–530, 1954.

[21] Vladimir I. Arnold. Proof of a theorem of A. N. Kolmogorov on the invariance of quasi-periodic motions under small perturbations of the Hamiltonian. *Russian Mathematical Surveys*, 18(5):9–36, 1963.

[22] Jürgen Moser. On invariant curves of area-preserving mappings of an annulus. *Nachrichten der Akademie der Wissenschaften in Göttingen, Mathematisch-Physikalische Klasse*, pages 1–20, 1962.

[23] Gregoire Nicolis and Ilya Prigogine. *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley-Interscience, New York, 1977.

[24] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002. URL https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

[25] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010. doi: 10.1152/jn.00106.2010.

[26] Martin Vinck, Robert Oostenveld, Marijn van Wingerden, Francesco Battaglia, and Cyriel M. A. Pennartz. An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias. *NeuroImage*, 55(4):1548–1565, 2011. doi: 10.1016/j.neuroimage.2011.01.055.

[27] David Tong. Lectures on electromagnetism. University of Cambridge lecture notes, 2015. URL https://www.damtp.cam.ac.uk/user/tong/em.html.

[28] National Institute of Standards and Technology. Meet the constants, 2018. URL https://www.nist.gov/si-redefinition/meet-constants. SI defining values for the Planck constant, elementary charge, and Boltzmann constant; accessed November 3, 2026.

[29] Andrea G. Markelz. Terahertz dielectric sensitivity to biomolecular structure and function. *IEEE Journal of Selected Topics in Quantum Electronics*, 14(1):180–190, 2008. doi: 10.1109/JSTQE.2007.913424.

[30] Herbert Fröhlich. Long-range coherence and energy storage in biological systems. *International Journal of Quantum Chemistry*, 2(5):641–649, 1968. doi: 10.1002/qua.560020505.